Linux Command Reference

Commands, key options, explanations and examples: CLI, permissions, users, processes, systemd, SSH

1 CLI & Files

Command	Key options	Explanation	Example / use case
pwd	-P (physical path, resolves symlinks)	Prints the current directory	`pwd -> /home/alice`
ls	-l long format; -a hidden files; -h human sizes; -t sort by time; -R recursive	Lists directory contents	`ls -lah /var/log (all logs with sizes and permissions)`
cd	~ home; - previous dir; .. parent	Changes directory	`cd /etc/nginx, then cd - to go back`
tree	-L n depth; -a hidden; -d dirs only	Shows a directory tree	`tree -L 2 /srv/project`
cp	-r recursive; -a archive (keeps perms/links); -i ask before overwrite; -v verbose	Copies files and directories	`cp -a /etc/nginx /backup/nginx-backup (back up a config before editing)`
mv	-i ask; -n never overwrite; -v verbose	Moves or renames	`mv app.log app.log.old (rename a log before rotating)`
rm	-r recursive; -f force; -i ask	Deletes files (no trash, no undo)	`rm -ri old_project/ (delete a folder with confirmations)`
mkdir	-p create parents; -m set mode	Creates directories	`mkdir -p /srv/project/{src,docs,logs}`
touch	-d "date"; -t timestamp	Creates an empty file or updates its timestamp	`touch notes.txt`
cat	-n number lines; -A show hidden chars	Prints file contents	`cat -n /etc/hosts`
less	/word search; G end; q quit; +F follow	Pages through large files	`less /var/log/syslog, then /error to search`
head	-n N first N lines	Shows the start of a file	`head -n 5 /etc/passwd`
tail	-n N; -f follow live; -F follow across rotation	Shows the end of a file	`tail -f /var/log/nginx/error.log (watch errors live)`
grep	-r recursive; -n line numbers; -i ignore case; -v invert; -c count; -E regex; -l filenames only	Searches text by pattern	`grep -rni "failed password" /var/log/auth.log (find SSH login failures)`
find	-name; -type f/d; -mtime +N; -size +10M; -user; -exec ... {} \;	Searches the filesystem by criteria	`find /var/log -name "*.log" -mtime +7 -exec rm {} \; (delete logs older than 7 days)`
man	-k keyword (search pages)	Opens the manual	`man chmod or man -k "copy files"`
--help	(flag on most commands)	Quick usage summary	`tar --help`

2 Pipes & Redirection

Command	Key options	Explanation	Example / use case
>	n/a	Redirect stdout, overwriting	`ls > files.txt`
>>	n/a	Redirect stdout, appending	`echo "done" >> deploy.log`

Command	Key options	Explanation	Example / use case
2>	n/a	Redirect errors only	`find / -name "*.conf" 2> /dev/null (hide "Permission denied" noise)`
2>&1	n/a	Merge errors into stdout	`./script.sh > out.log 2>&1 (capture everything in one file)`
|	n/a	Send one command's output to another	`ps aux | grep nginx (find nginx processes)`

3 Permissions

Command	Key options	Explanation	Example / use case
ls -l	n/a	Shows -rwxr-xr--: type, owner, group, others. r=4, w=2, x=1	`ls -l script.sh`
chmod	Numeric (750) or symbolic (u+x, g-w, o=r); -R recursive	Changes permissions	`chmod 750 deploy.sh (owner rwx, group r-x, others none); chmod u+x script.sh (make executable)`
chmod special bits	4xxx setuid; 2xxx setgid; 1xxx sticky	setuid runs as file owner; setgid makes new files inherit the group; sticky means only owners can delete files in the dir	`chmod 2770 /srv/project (shared team folder); chmod 1777 /tmp (standard sticky setup)`
chown	user:group; -R recursive	Changes owner and group	`chown -R deploy:devs /srv/project`
chgrp	-R recursive	Changes group only	`chgrp devs report.txt`
umask	022; 027	Sets the default permission mask for new files (files start at 666, dirs at 777, minus the mask)	`umask 027 (new files 640, new dirs 750)`

4 Users & Groups

Command	Key options	Explanation	Example / use case
/etc/passwd, /etc/shadow, /etc/group	n/a	Account info, password hashes (root only), group membership	`getent passwd alice`
useradd	-m create home; -s shell; -G extra groups; -u UID; -c comment	Creates a user	`useradd -m -s /bin/bash -G devs deploy`
passwd	-l lock; -u unlock; -e force change at next login	Sets or manages passwords	`passwd deploy`
usermod	-aG append to groups; -s shell; -L lock; -U unlock; -l rename	Modifies a user	`usermod -aG sudo alice (always use -a, or you replace all existing groups)`
userdel	-r also remove home and mail	Deletes a user	`userdel -r alice`
groupadd	-g GID	Creates a group	`groupadd devs`
groupdel	n/a	Deletes a group	`groupdel devs`
id	-u; -g; -Gn	Shows UID, GID, and groups	`id alice`
groups	n/a	Lists a user's groups	`groups deploy`
whoami	n/a	Prints the current user	`whoami -> alice`

Command	Key options	Explanation	Example / use case
sudo	-u user run as user; -l list rights; -i root shell	Runs a command with elevated privileges	`sudo systemctl restart nginx`
su	- (login shell); -c cmd	Switches user	`su - deploy`
visudo	-c check syntax	Safely edits /etc/sudoers	`visudo, then add: alice ALL=(ALL) NOPASSWD: /usr/bin/systemctl`

5 Processes & Resources

Command	Key options	Explanation	Example / use case
ps	aux all processes; -ef full format; -u user; --forest tree	Snapshot of processes	`ps aux --sort=-%mem | head (top memory consumers)`
top	-u user; keys: M sort by memory, P by CPU, k kill, q quit	Live process monitor	`top -u www-data`
htop	F4 filter; F9 kill; F6 sort	Friendlier interactive monitor (may need install)	`htop`
pgrep	-l show names; -u user; -f match full command	Finds PIDs by name	`pgrep -l nginx`
kill	-15 SIGTERM (graceful, default); -9 SIGKILL (force); -HUP reload; -l list signals	Sends a signal to a PID	`kill 1234; if ignored, kill -9 1234`
killall	-9; -u user	Kills by process name	`killall firefox`
pkill	-f; -u user	Kills by pattern	`pkill -f "python app.py"`
& / jobs / fg / bg	fg %1; bg %1	Run in background, list jobs, bring to front, resume in background	`sleep 1000 &, then jobs, then fg %1`
nohup	nohup cmd &	Keeps a process running after logout	`nohup ./backup.sh > backup.log 2>&1 &`
nice / renice	-n 10 (lower priority); -p PID. Range: -20 highest, 19 lowest	Sets or changes priority	`nice -n 10 ./heavy_job.sh; renice 15 -p 1234`
free	-h human; -m MB	Shows memory and swap	`free -h`
df	-h human; -T filesystem type	Shows disk usage per filesystem	`df -h`
du	-sh summary; --max-depth=1	Shows directory sizes	`du -sh /var/log/* (find what fills the disk)`
uptime	-p pretty	Shows uptime and load average	`uptime`
lsof	-i :80 by port; -u user; -p PID	Lists open files and sockets	`lsof -i :80 (who uses port 80)`
ss	-tulpn listening TCP/UDP with process	Shows sockets	`ss -tulpn (which services are listening)`

6 Services (systemd)

Command	Key options	Explanation	Example / use case
systemctl status	-l full lines	Shows state and recent logs of a service	`systemctl status nginx`
systemctl start / stop / restart	n/a	Starts, stops, or restarts now	`systemctl restart nginx (after a config change)`
systemctl reload	n/a	Reloads config without dropping connections	`systemctl reload nginx`

Command	Key options	Explanation	Example / use case
systemctl enable / disable	--now also start or stop immediately	Controls start at boot	`systemctl enable --now nginx`
systemctl is-active / is-enabled	n/a	Scriptable state checks	`systemctl is-active sshd`
systemctl list-units	--type=service; --state=failed	Lists units	`systemctl list-units --type=service --state=failed`
systemctl mask / unmask	n/a	Completely prevents a service from starting	`systemctl mask bluetooth`
systemctl daemon-reload	n/a	Re-reads unit files after you edit them	`Run after editing /etc/systemd/system/myapp.service`
journalctl	-u unit; -f follow; -xe recent errors with help; --since "1 hour ago"; -p err; -b current boot	Reads systemd logs	`journalctl -u nginx --since "10 min ago" -p err (diagnose a broken config)`

7 SSH & Keys

Command	Key options	Explanation	Example / use case
ssh-keygen	-t ed25519 type; -b 4096 (RSA size); -C comment; -f file path	Generates a key pair	`ssh-keygen -t ed25519 -C "you@example.com" -f ~/.ssh/id_deploy`
ssh-copy-id	-i keyfile; -p port	Installs your public key on a server	`ssh-copy-id -i ~/.ssh/id_deploy.pub deploy@192.168.1.10`
ssh	-i key; -p port; -v debug; -L local port forward; -J jump host	Connects to a remote machine	`ssh -i ~/.ssh/id_deploy -p 2222 deploy@server`
scp	-r recursive; -P port; -i key	Copies files over SSH	`scp -r ./site deploy@server:/var/www/`
ssh-agent / ssh-add	ssh-add -l list keys	Holds decrypted keys so you type the passphrase once	`eval "$(ssh-agent)" && ssh-add ~/.ssh/id_deploy`
~/.ssh/config	Host; HostName; User; Port; IdentityFile	Client shortcuts	`Host prod / HostName 192.168.1.10 / User deploy / Port 2222 / IdentityFile ~/.ssh/id_deploy -> then just: ssh prod`
~/.ssh/authorized_keys	One public key per line	Server-side list of allowed keys	`cat id_deploy.pub >> ~/.ssh/authorized_keys`
/etc/ssh/sshd_config	PasswordAuthentication no; PermitRootLogin no; Port 2222; AllowUsers deploy	Server hardening	`Edit, run sshd -t to validate, then systemctl restart sshd`
SSH permissions	~/.ssh = 700; private key and authorized_keys = 600	SSH ignores keys if permissions are too open	`chmod 700 ~/.ssh && chmod 600 ~/.ssh/id_ed25519 ~/.ssh/authorized_keys`

Bonus: minimal systemd unit file

`/etc/systemd/system/myapp.service` : then run `systemctl daemon-reload && systemctl enable --now myapp`

```
[Unit]
Description=My App
After=network.target

[Service]
User=deploy
ExecStart=/usr/bin/python3 /srv/project/app.py
Restart=on-failure

[Install]
WantedBy=multi-user.target
```